Lab Assignment Instructions

Dr. Frank Xu

October 23, 2026

Objective

This assignment guides you through the commands in the lecture slides. You will execute each required command, capture its relevant output, explain what the command and result mean, and document any errors you encounter.

Instructions

Use the following format for every submission:

- Standard 8.5-by-11-inch pages.
- At least 12-point text throughout, including text in figures and screenshots.

Step 1: Access the Lab Instructions

1. Open the lecture slides and lab resources in the GitHub repository.
2. Download the assigned lecture slides. Figure 1 shows the download process.
3. Review the slides and identify the required tasks and their order.

Step 2: Complete the Assigned Tasks

1. Complete the tasks in the order shown in the lecture slides.
2. Execute each required command and verify its result.

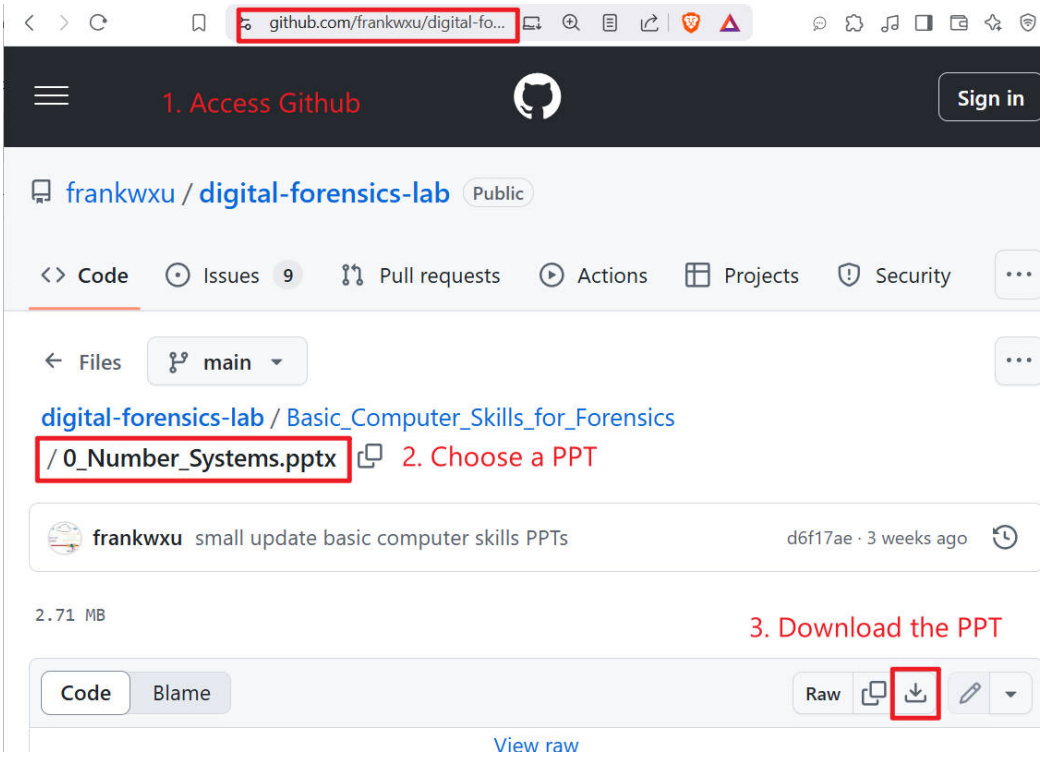


Figure 1: Steps for downloading the lecture slides

Step 3: Capture Screenshots

1. Capture the relevant output of each required command.
2. Crop each screenshot to the relevant command and output. Remove unnecessary blank space and errors unrelated to the task.
3. Use screenshots with a resolution at least as high as the example in Figure 2.
4. In the submitted document, all text visible in a screenshot must appear at least as large as 12-point text.
5. Label each screenshot with its corresponding task or step number.
6. You may use Snipaste (`https://www.snipaste.com/`) or the Microsoft Snipping Tool to capture and crop screenshots.

Step 4: Explain Your Work

1. For each command, explain its purpose, options or arguments, and result. Use at least 12-point text.

2. Follow this example:

   Command: `ll *.txt`

   Description: In the `~/readme` directory, `ll` displays a long listing when the system defines it as an alias for `ls -l`. The `*.txt` wildcard limits the results to names ending in `.txt`. Each row shows the file's permissions, link count, owner, group, size, modification date and time, and name.

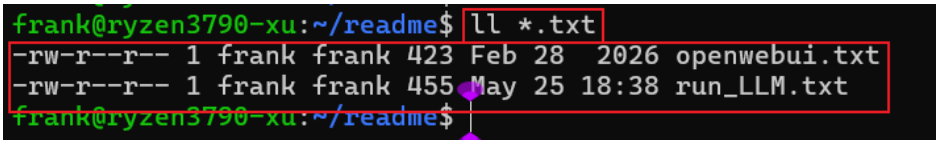


Figure 2: A sample screenshot of the command `ll *.txt`

   Result: The output in Figure 2 lists the two matching text files, `openwebui.txt` and `run_LLM.txt`. The leading hyphen in each permission string indicates a regular file. Both files are owned by user `frank` and group `frank`; the remaining columns show their sizes and last modification dates and times.

Step 5: Handle Errors

1. Record errors that prevent you from completing an assigned task.

2. Include a cropped screenshot showing the relevant error message.

3. Briefly describe the error and the troubleshooting steps you tried.

4. Errors caused by operating-system or software-version differences will not affect your grade when they are documented as directed.

Submission Guidelines

- Combine all screenshots, explanations, and error documentation into one PDF.
- Name the file `LastName_FirstName_LabAssignment_XXX.pdf`.
- Submit the PDF through Canvas by the stated deadline (one week by default).

Grading Criteria

Each lab is worth 10 points unless otherwise announced. A lab without a hands-on component will use an alternative 10-point problem-solving assignment. **Note:** Screenshots and command descriptions in the lecture slides are teaching examples and may not satisfy the criteria below. Follow this document when preparing your submission.

- **Completion of all tasks (50%)**
  - Missing required tasks or steps.
  - Missing required commands or results.
  - Evidence that does not demonstrate successful completion.
- **Accuracy of explanations (30%)**
  - Missing explanations.
  - Inaccurate or vague explanations. For example, "`ll *.txt` shows files" is too vague because it does not explain the long-listing format or the wildcard. "`ll *.txt` lists every file in the directory" is inaccurate because the command lists only names ending in `.txt`.
  - Repeating the command without explaining its purpose, options, or result.
- **Quality of screenshots and documentation (20%)**
  - Text smaller than 12 points in figures or screenshots.
  - Figures or screenshots with lower resolution than the example.
  - Screenshots with unrelated commands, output, or errors.
  - Commands or results that are not clearly highlighted.
  - Unnecessary blank space or screenshots not cropped to the relevant command and output.

Additional Notes

- Ask the instructor or teaching assistant about course content.
- Contact the Office of Technology Services (OTS) for Canvas issues.
- Save your work frequently to avoid data loss.